BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 277 · Normalisation to 3NF and indexing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a transitive dependency?

Q2. What does an index help with?

Q3. What does 1NF require?

Q4. Simplify (A∧B)∨(A∧¬B), stating the Boolean law.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

1NF — Atomic values/no repeating groups.

2NF — No partial dependency on part of composite key.

3NF — No non-key transitive dependency violating key dependency.

index — Search structure over columns.

New learning

Understand — 1NF requires atomic values and no repeating groups in a record; 2NF removes dependencies on only part of a composite candidate key.

Apply — 3NF eliminates inappropriate transitive dependencies among non-key facts, reducing repeated customer names across orders.

Evaluate — An index accelerates some lookups but uses storage and may slow insert/update operations. Normalisation and indexing solve different problems.

Worked example

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

1. Predict first — Store customer name once in Customer and reference by CustomerID in Order.

2. Model answer — Explain the reasoning and justify the output.

We Do · Practise with guidance

SalesLine(OrderID,ProductID,ProductName,Quantity). Remove product-name dependency.

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Product(ProductID,ProductName)
SalesLine(OrderID,ProductID,Quantity)
```

Predict first

ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.

Model answer

Explain the reasoning and justify the output.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State 1NF,2NF,3NF in one sentence each. [6]

Question 2

Order(OrderID, CustomerID, CustomerName). Explain the redundancy and split into two tables. [6]

Question 3

Explain why an index on StudentName can speed SELECT but also incur costs. [6]

Exit Ticket · Answer before checking

Normalising to 3NF means the database never needs indexes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. An attribute depends on key through another non-key attribute.

Q2. Search speed on indexed fields.

Q3. Atomic fields and no repeating groups.

Q4. $A \wedge (B \vee \neg B) = A \wedge 1 = A$.

You Do pages · Independent questions

Q1. 1NF atomic/no repeating groups [2]; 2NF removes partial dependencies on composite key [2]; 3NF removes problematic transitive dependencies [2].

Q2. CustomerName repeated for one CustomerID [2]; Customer(CustomerID PK, CustomerName) [2]; Order(OrderID PK,CustomerID FK) [2].

Q3. Index helps locate selected names [2], can reduce scanning [1]; index uses storage [1], maintenance on writes [1], duplicates may need extra row matching [1].

Exit Ticket · Model answer

False. Normalisation and indexing have different purposes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: